

WALEED IQBAL

Senior Software Engineer | Agentic AI Evaluation and Agent Infrastructure

Faisalabad, Pakistan | Open to relocation with visa sponsorship | Available remote across US and EU hours

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Portfolio: waleediqbal.vercel.app

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PROFESSIONAL SUMMARY

Senior software engineer specialising in agentic AI evaluation infrastructure: the systems that measure whether AI agents actually work. Provider agnostic execution across frontier and self hosted models, reproducible sandboxed browser and desktop environments, layered automated grading, and the distributed orchestration that keeps long running agent runs healthy. Five years across production SaaS delivery and AI infrastructure, promoted from associate to senior before moving into evaluation platform work.

TECHNICAL SKILLS

Agent systems and evaluation: computer use agents, multi model evaluation, LLM as judge, rubric design, trajectory analysis, Playwright, OSWorld, WebArena, BrowseComp

Backend and orchestration: Python, FastAPI, Django, Celery, Redis, PostgreSQL, REST APIs, distributed systems, Ruby on Rails, Node.js

Cloud and infrastructure: GCP, AWS, Kubernetes, Docker, QEMU/KVM, Prometheus, GitLab CI/CD

Frontend and tooling: React, Next.js, TypeScript, Electron, Chrome extensions, FFmpeg, WebRTC, vLLM

PROFESSIONAL EXPERIENCE

Turing | Remote, full time contract | Sep 2023 - Present

Software Engineer, Agentic AI Evaluation Platform | Sep 2025 - Present

- Brought the evaluation platform from nothing to production in three months on FastAPI, Celery, Redis, PostgreSQL and Next.js across GCP and AWS. Now the daily tool for around 500 people spanning engineering, task authoring, trajectory review and client delivery.
- Built the provider agnostic execution layer that drives OpenAI, Anthropic, Gemini and self hosted checkpoints on vLLM through one loop, normalising divergent tool calling formats, action vocabularies, coordinate conventions and token accounting into a single trajectory format.
- Designed the layered verification system behind grading: golden trajectory comparison, LLM evaluators, DOM and state assertions, and graders executing inside the target environment. Single signal grading drifts between runs and fails an agent that solved the task by a different valid route.
- Sustained 30+ concurrent long running agent tasks with zero duplicate dispatch and automatic recovery from worker loss, using atomic database claims, PostgreSQL advisory locks, Redis leases and watchdog sweeps across a Celery worker fleet.
- Made every delivery traceable to a signed off task version, with per run model spend and full audit events behind it, so a question about how a number was produced is answered from the record rather than from memory.

Software Engineer, Agentic RL Environments | Jul 2024 - Aug 2025

- Built reinforcement learning web environments by rebuilding production SaaS applications as self contained offline React apps with high fidelity state management, so agent behaviour could be measured deterministically across CRM, commerce, support desk and booking domains.
- Authored multi step agent workflows and their grader logic, replacing ambiguous answer based grading with deterministic state based grading so the evaluation signal could be trusted.
- Put browser and full desktop runs on one execution path across managed Kubernetes and cloud VMs, covering OSWorld, WebArena and BrowseComp workloads, with desktops booting from prebuilt images for reproducible startup.

Software Engineer, LLM Code Generation and Computer Use | Sep 2023 - Jun 2024

- Authored and evaluated Python implementations used to train and benchmark LLM code generation, judged on correctness, algorithmic efficiency and code quality, progressing from individual algorithmic problems to complete multi iteration Django and React applications.
- Validated end to end development workflows across macOS, Ubuntu and Chromium, using prompt engineering to expose model failure modes in code comprehension, iterative programming and debugging.

Softaims | Lahore, Pakistan | Aug 2021 - Jul 2024 | promoted each year

Senior Software Engineer | Aug 2023 - Jul 2024

Held concurrently with the Turing contract, with both employers' knowledge.

- Directed architecture, delivery planning and release decisions across three concurrent client products with a team of three engineers, reporting production risk directly to the clients themselves.

- Cleared Shopify App Store review with an upsell app rebuilt from a five year old codebase: theme app extensions, recommendations driven by order history and stated customer demand, and a GraphQL rewrite that carried it through Shopify's API transition.

Software Engineer | Aug 2022 - Aug 2023

- Handed release control to a non technical client running a large network of German institutes: every merged branch deployed itself to a review environment he signed off on, then reached production on a button click, by rebuilding delivery on GitLab CI/CD with Dockerised builds shipping to Heroku.
- Let every institute on that platform run under its own subdomain on a shared Rails and React codebase, and rebuilt the interface on Tailwind while integrating the APIs the institutes are administered through.
- Built a Shopify app with Stripe billing and webhook handling, including the idempotency and replay handling that keeps a billing webhook from charging twice when the network retries it.

Associate Software Engineer | Aug 2021 - Aug 2022

- Gave support agents live sight of a customer's screen through an embeddable WebRTC tool with video and audio calling, plus a companion Chrome extension recording captures straight to storage in three formats via FFmpeg.
- Dropped latency on a crypto platform buckling under server load by moving realtime third party data fetching onto Redis backed workers with caching, so the main backend service stopped absorbing traffic it could not carry.

OPEN SOURCE PROJECTS

Three deployed, tested repositories that together form a working agent evaluation stack. Written up at waleediqbal.vercel.app.

agentscore — evaluation harness for computer use agents

Next.js, TypeScript, Playwright, Python | Live: agentscore-sigma.vercel.app | Code: github.com/iamwaleediqbal/agentscore

- Drives a live web application in a real browser and grades the world before the task against the world after it, never the route taken. Both snapshots stay on the run record, so a change to the grading logic is retested against all 48 past runs at no further model cost. 48 runs across 5 models, 6 tasks and 2 action spaces, for \$0.65.
- Scores each action space separately rather than blending them: three of the five models scored higher reading the screen as text than as pixels, one at 6/6 against 3/6 on identical tasks. Reported as intervals, with overlapping ones declared ties.

polyact — one action schema across model providers

Python | Code: github.com/iamwaleediqbal/polyact

- Normalises four incompatible provider response formats into one action record in real screen pixels. Gemini answers on a 0-1000 grid where OpenAI answers in screenshot pixels, so reading one as the other puts every click in the top left quarter of the screen and scores a capable model as blind, with no error raised anywhere.
- Token accounting that survives a provider change: OpenAI counts cached tokens inside its input total and Anthropic does not, so a cached run reads cheaper than it was until the invoice arrives.
- Publishes the coordinate rules as machine readable conformance vectors, so a second implementation in another language can be held to them. agentscore's TypeScript runner is tested against the same file and goes red if the two drift.

clickmail — the application under test

Next.js, TypeScript | Live: clickmail-sigma.vercel.app/gym | Code: github.com/iamwaleediqbal/clickmail

- A public mail client with 52 messages across 7 folders, search, labels, spam and a composer, built to be operated by an agent rather than a person. It holds no grader, no tasks and no models, because an environment that contains its own grader can only ever score itself.
- Publishes a three method automation contract: report the world, reset it, and list the controls currently on screen so a harness can refuse to spend a turn on an action the interface no longer offers. There is deliberately no way to install a state from outside, because a driver that could write the world could write the answer.

SELECTED PROJECTS

- **AI chatbot platform for Shopify merchants** (Node.js, TypeScript, Shopify Remix, Prisma, PostgreSQL with pgvector, BullMQ, OpenRouter): merchants connect a store over OAuth and a one time quiz generates a per store system prompt. The assistant answers shoppers through real time tool calls against store data rather than bulk context injection, with pgvector retrieval and order lookup gated by order number and email verification. Working build, store listing in preparation.

EDUCATION

BSc Computer Science | FAST National University of Computer and Emerging Sciences, Lahore, Pakistan | 2017 - 2021

LANGUAGES

English (professional working proficiency) | Urdu (native) | Punjabi (native) | Hindi (spoken, not written)